

DBI Cheatsheet — R + Databases in 7 Functions

Connect, query, disconnect

```
library(DBI)
con <- dbConnect(RSQLite::SQLite(), ":memory:") # or duckdb::duckdb()
dbListTables(con)                             # what is in the database?
dbReadTable(con, "ecom")                      # whole table (small only!)
dbGetQuery(con, "SELECT * FROM ecom LIMIT 10") # SQL, first rows
res <- dbSendQuery(con, "SELECT * FROM ecom")  # big tables: fetch in batches
x <- dbFetch(res, n = 100)
dbClearResult(res)                            # always clear when done
dbDisconnect(con)                             # always disconnect
```

Write data

```
dbWriteTable(con, "trial", df)                # create (overwrite / append args)
dbExecute(con, "INSERT INTO trial (x) VALUES (1)") # statements, no cleanup needed
dbRemoveTable(con, "trial")                   # delete table
```

Inspect a query

dbHasCompleted() dbGetStatement() dbGetRowCount() dbGetRowsAffected() dbColumnInfo() — status, SQL, counts, column types of a dbSendQuery() result.

dbplyr in 4 verbs

```
e <- dplyr::tbl(con, "ecom")                  # lazy reference, pulls nothing
e %>% dplyr::filter(duration > 300)           # WHERE
e %>% dplyr::group_by(device) %>%             # GROUP BY + aggregates
  dplyr::summarise(avg = mean(duration))
dplyr::show_query(q)                          # see the generated SQL
dplyr::collect(q)                             # pull results into a tibble
inner_join(a, b) / left_join(a, b)            # JOINS; semi/anti_join filter
```